

Kahoot!

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OBJECTIVE

- A common issue we found, is Elderly people with high blood pressure, forgetting to take their medication, which later leads to further complications with their health. A study from the National Library of Medicine, further backs up this finding, by showing that on average only 20-50% of individuals with Hypertension, adhere to taking their medication contributing to the 670,000 deaths from the condition in 2020. (Ruksakulpiwat et al.)
- Our project is simple, but effective. Our device reminds users through the device's vibration, to take their medication. Once they press a button on the device, it dispenses the pill and stops the alert. If not, a safety contact is notified of the missed medication and is then urged to ensure compliance. This way, there is a backup safeguard to make sure the medication is taken.

KPP AND TPM

- To ensure that the device would be effective at solving the issue, we defined a few KPP's as well as some TPM to measure the overall success.

The device should be wrist-based or wearable

The device should notify the user via haptic vibration at the set time

The device should require explicit button press to acknowledge medication taken

The device should alert a safety contact if button is not pressed in time

- When measuring success based on Technical Parameter:

The shell measures no greater than 95 mm (L) × 75 mm (W) × 25 mm (H).

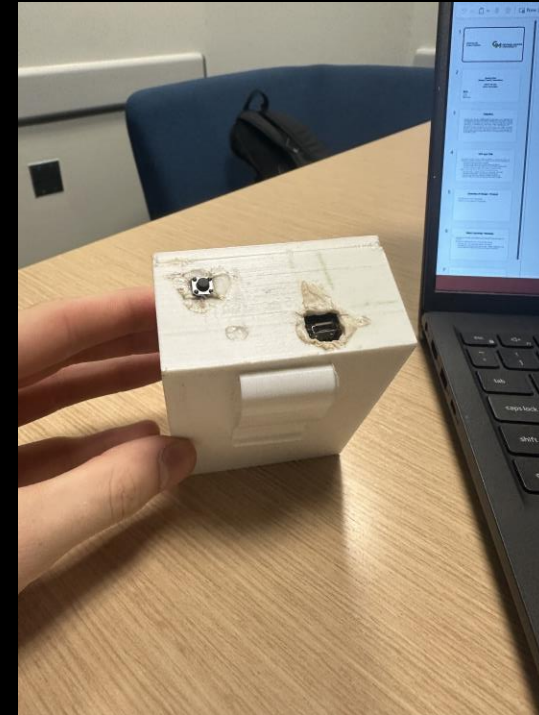
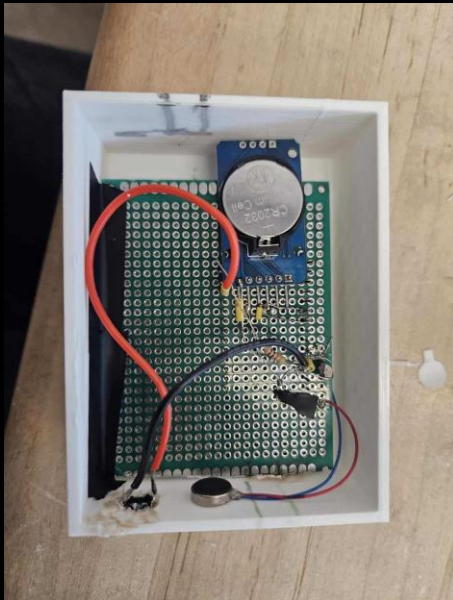
Total system weight (device + strap) must not exceed 150 grams.

The alert triggers within 15 seconds of the scheduled medication time.

Battery life supports a minimum of 5 days of operation between charges.

OVERVIEW OF DESIGN / PRODUCT

- Provide pictures of your final product
- Include any circuit diagrams (if applicable)



MAJOR LEARNINGS TAKEAWAY

- **Waterfall Process Reflection**
- Helped create a feasible, well-planned product
- Designing before implementation was critical to avoid getting stuck later
- However, the requirements phase felt limiting and slowed progress
- **Project Outcome**
- Developed a clip-on medical alert device
- Very satisfied — resulted in a physical, working product
- Future improvement: optimize and refine the product design
- **Challenges**
- Making a small device with limited electrical engineering skills.
- **Were milestones effective?**
- Helped guide us through waterfall process

SOURCE

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC11088410/>